```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Investment Data Analysis Project

```
df = pd.read_csv("Data_set 2 - Copy (1).csv")
```

```
print(df.head())
```

```
   gender  age Investment_Avenues  Mutual_Funds  Equity_Market  Debentures  \
0  Female   34                Yes             1              2           5
1  Female   23                Yes             4              3           2
2    Male   30                Yes             3              6           4
3    Male   22                Yes             2              1           3
4  Female   24                 No             2              1           3

   Government_Bonds  Fixed_Deposits  PPF  Gold  ...           Duration  \
0                 3               7    6     4  ...          1-3 years
1                 1               5    6     7  ...  More than 5 years
2                 2               5    1     7  ...          3-5 years
3                 7               6    4     5  ...   Less than 1 year
4                 6               4    5     7  ...   Less than 1 year

  Invest_Monitor   Expect       Avenue What are your savings objectives?  \
0        Monthly  20%-30%  Mutual Fund                   Retirement Plan
1         Weekly  20%-30%  Mutual Fund                       Health Care
2          Daily  20%-30%       Equity                   Retirement Plan
3          Daily  10%-20%       Equity                   Retirement Plan
4          Daily  20%-30%       Equity                   Retirement Plan

          Reason_Equity         Reason_Mutual     Reason_Bonds  \
0  Capital Appreciation        Better Returns  Safe Investment
1              Dividend        Better Returns  Safe Investment
2  Capital Appreciation          Tax Benefits  Assured Returns
3              Dividend  Fund Diversification   Tax Incentives
4  Capital Appreciation        Better Returns  Safe Investment

             Reason_FD                    Source
0        Fixed Returns  Newspapers and Magazines
1  High Interest Rates     Financial Consultants
2        Fixed Returns                Television
3  High Interest Rates                  Internet
4            Risk Free                  Internet

[5 rows x 24 columns]
```

Task 1: Data Overview

```
#task 1

print(df.info())
print(df.describe())
print(df.shape)
print(df.columns)
```

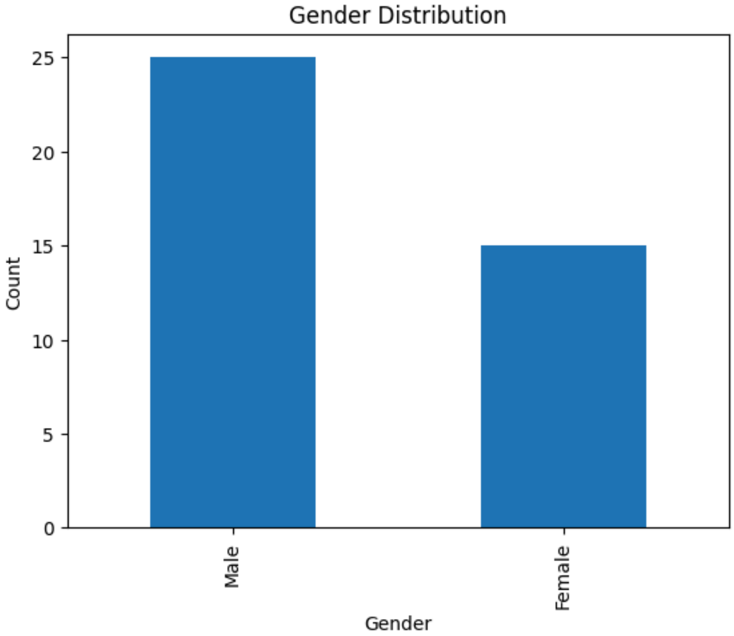
```
RangeIndex: 40 entries, 0 to 39
Data columns (total 24 columns):
 #   Column                             Non-Null Count  Dtype
---  ------                             --------------  -----
 0   gender                             40 non-null     object
 1   age                                40 non-null     int64
 2   Investment_Avenues                 40 non-null     object
 3   Mutual_Funds                       40 non-null     int64
 4   Equity_Market                      40 non-null     int64
 5   Debentures                         40 non-null     int64
 6   Government_Bonds                   40 non-null     int64
 7   Fixed_Deposits                     40 non-null     int64
 8   PPF                                40 non-null     int64
 9   Gold                               40 non-null     int64
 10  Stock_Marktet                      40 non-null     object
 11  Factor                             40 non-null     object
 12  Objective                          40 non-null     object
 13  Purpose                            40 non-null     object
 14  Duration                           40 non-null     object
 15  Invest_Monitor                     40 non-null     object
 16  Expect                             40 non-null     object
 17  Avenue                             40 non-null     object
 18  What are your savings objectives?  40 non-null     object
```

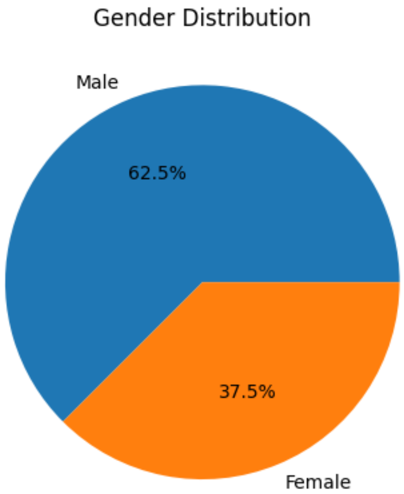
```
 20  Reason_Mutual                       40 non-null     object
 21  Reason_Bonds                        40 non-null     object
 22  Reason_FD                           40 non-null     object
 23  Source                              40 non-null     object
dtypes: int64(8), object(16)
memory usage: 7.6+ KB
None
             age  Mutual_Funds  Equity_Market  Debentures  Government_Bonds  \
count  40.000000     40.000000      40.000000   40.000000         40.000000
mean   27.800000      2.550000       3.475000    5.750000          4.650000
std     3.560467      1.197219       1.131994    1.675617          1.369072
min    21.000000      1.000000       1.000000    1.000000          1.000000
25%    25.750000      2.000000       3.000000    5.000000          4.000000
50%    27.000000      2.000000       4.000000    6.500000          5.000000
75%    30.000000      3.000000       4.000000    7.000000          5.000000
max    35.000000      7.000000       6.000000    7.000000          7.000000

       Fixed_Deposits        PPF       Gold
count       40.000000  40.000000  40.000000
mean         3.575000   2.025000   5.975000
std          1.795828   1.609069   1.143263
min          1.000000   1.000000   2.000000
25%          2.750000   1.000000   6.000000
50%          3.500000   1.000000   6.000000
75%          5.000000   2.250000   7.000000
max          7.000000   6.000000   7.000000
(40, 24)
Index(['gender', 'age', 'Investment_Avenues', 'Mutual_Funds', 'Equity_Market',
       'Debentures', 'Government_Bonds', 'Fixed_Deposits', 'PPF', 'Gold',
       'Stock_Marktet', 'Factor', 'Objective', 'Purpose', 'Duration',
       'Invest_Monitor', 'Expect', 'Avenue',
       'What are your savings objectives?', 'Reason_Equity', 'Reason_Mutual',
       'Reason_Bonds', 'Reason_FD', 'Source'],
      dtype='object')
```

Task 2: Gender Distribution

```
#task 2

print(df.columns)

gender_count = df['gender'].value_counts()
print(gender_count)

gender_count.plot(kind='bar')
plt.title("Gender Distribution")
plt.xlabel("Gender")
plt.ylabel("Count")
plt.show()

gender_count.plot(kind='pie', autopct='%1.1f%%')
plt.title("Gender Distribution")
plt.ylabel("")
plt.show()
```

```
Index(['gender', 'age', 'Investment_Avenues', 'Mutual_Funds', 'Equity_Market',
       'Debentures', 'Government_Bonds', 'Fixed_Deposits', 'PPF', 'Gold',
       'Stock_Marktet', 'Factor', 'Objective', 'Purpose', 'Duration',
       'Invest_Monitor', 'Expect', 'Avenue',
       'What are your savings objectives?', 'Reason_Equity', 'Reason_Mutual',
       'Reason_Bonds', 'Reason_FD', 'Source'],
      dtype='object')
gender
Male      25
Female    15
Name: count, dtype: int64
```





Task 3: Descriptive Statistics

```
#task 3

num_cols = df.select_dtypes(include=['int64', 'float64'])
print(num_cols.columns)

print(num_cols.mean())

print(num_cols.median())

print(num_cols.std())

print(num_cols.describe())
```

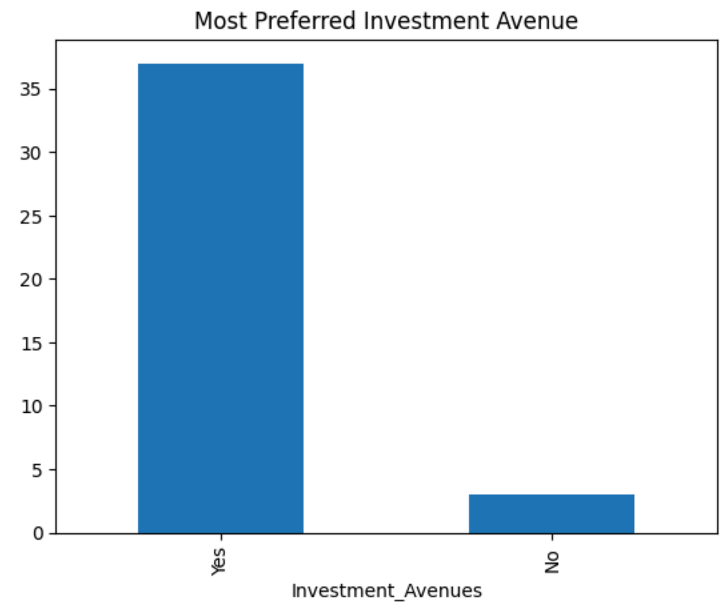
```
Index(['age', 'Mutual_Funds', 'Equity_Market', 'Debentures',
       'Government_Bonds', 'Fixed_Deposits', 'PPF', 'Gold'],
      dtype='object')
age                 27.800
Mutual_Funds         2.550
Equity_Market        3.475
```

```
Debentures          5.750
Government_Bonds    4.650
Fixed_Deposits      3.575
PPF                 2.025
Gold                5.975
dtype: float64
age                27.0
Mutual_Funds        2.0
Equity_Market       4.0
Debentures          6.5
Government_Bonds    5.0
Fixed_Deposits      3.5
PPF                 1.0
Gold                6.0
dtype: float64
age                3.560467
Mutual_Funds       1.197219
Equity_Market      1.131994
Debentures         1.675617
Government_Bonds   1.369072
Fixed_Deposits     1.795828
PPF                1.609069
Gold               1.143263
dtype: float64
             age  Mutual_Funds  Equity_Market  Debentures  Government_Bonds  \
count  40.000000     40.000000      40.000000   40.000000         40.000000
mean   27.800000      2.550000       3.475000    5.750000          4.650000
std     3.560467      1.197219       1.131994    1.675617          1.369072
min    21.000000      1.000000       1.000000    1.000000          1.000000
25%    25.750000      2.000000       3.000000    5.000000          4.000000
50%    27.000000      2.000000       4.000000    6.500000          5.000000
75%    30.000000      3.000000       4.000000    7.000000          5.000000
max    35.000000      7.000000       6.000000    7.000000          7.000000

       Fixed_Deposits        PPF       Gold
count       40.000000  40.000000  40.000000
mean         3.575000   2.025000   5.975000
std          1.795828   1.609069   1.143263
min          1.000000   1.000000   2.000000
25%          2.750000   1.000000   6.000000
50%          3.500000   1.000000   6.000000
75%          5.000000   2.250000   7.000000
max          7.000000   6.000000   7.000000
```

Task 4: Most Preferred Investment Avenue

```
#task 4

print(df['Investment_Avenues'].value_counts())

df['Investment_Avenues'].value_counts().plot(kind='bar')
plt.title("Most Preferred Investment Avenue")
plt.show()
```

```
Investment_Avenues
Yes    37
No      3
Name: count, dtype: int64
```

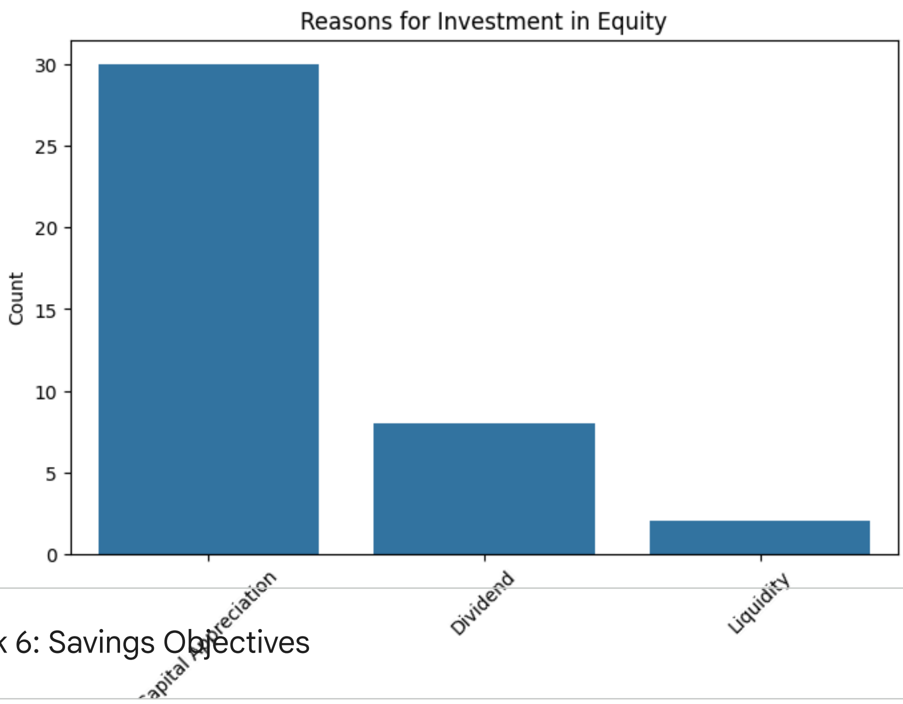


```
import pandas as pd
df = pd.read_csv("Data_set 2 - Copy (1).csv")
print(df.columns)
```

```
Index(['gender', 'age', 'Investment_Avenues', 'Mutual_Funds', 'Equity_Market',
       'Debentures', 'Government_Bonds', 'Fixed_Deposits', 'PPF', 'Gold',
       'Stock_Marktet', 'Factor', 'Objective', 'Purpose', 'Duration',
       'Invest_Monitor', 'Expect', 'Avenue',
       'What are your savings objectives?', 'Reason_Equity', 'Reason_Mutual',
       'Reason_Bonds', 'Reason_FD', 'Source'],
      dtype='object')
```

Task 5: Reasons for Investment

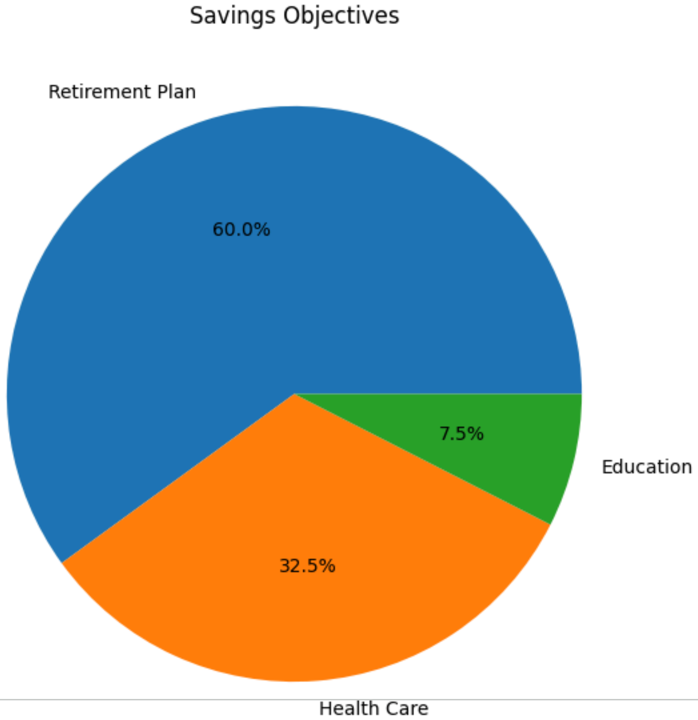
```
#task 5

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("Data_set 2 - Copy (1).csv")

reasons = df['Reason_Equity'].value_counts() # Changed 'YOUR_COLUMN_NAME' to 'Reason_Equity' as an example.

plt.figure(figsize=(8,5))
sns.barplot(x=reasons.index, y=reasons.values)

plt.title("Reasons for Investment in Equity") # Updated title to reflect the column used
plt.xlabel("Reasons")
plt.ylabel("Count")
plt.xticks(rotation=45)

plt.show()
```



Task 6: Savings Objectives

```
#task 6

'What are your savings objectives?'
print(df['What are your savings objectives?'].value_counts())

df['What are your savings objectives?'].value_counts().plot(
    kind='pie',
    autopct='%1.1f%%',
    figsize=(7,7)
)

plt.title("Savings Objectives")
plt.ylabel("")

plt.show()
```

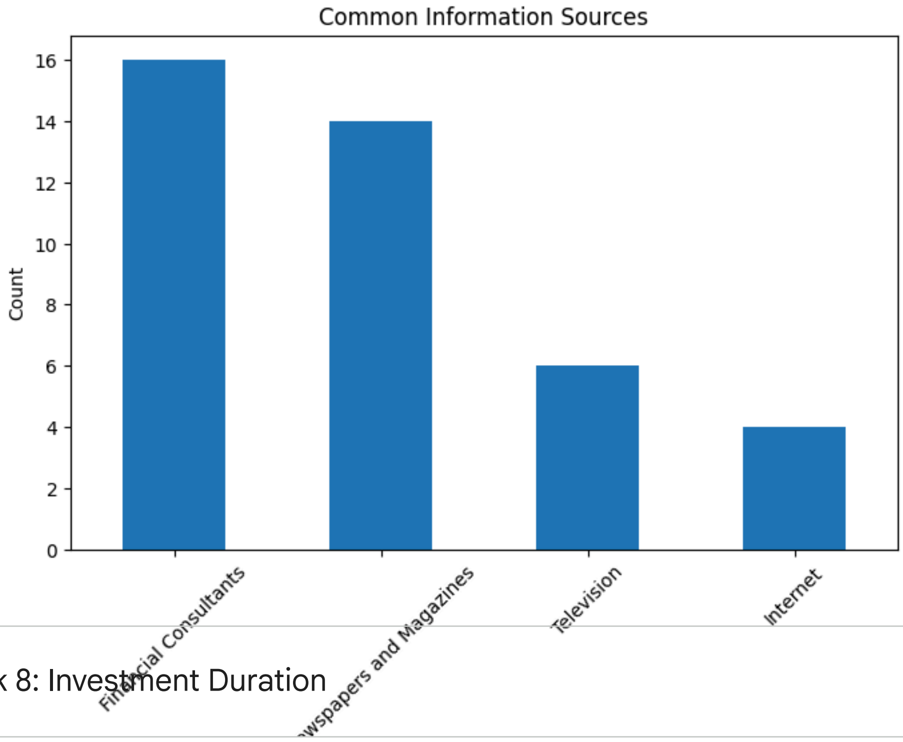
```
What are your savings objectives?
Retirement Plan    24
Health Care        13
Education           3
Name: count, dtype: int64
```



Task 7: Common Information Sources

```
#task 7

print(df['Source'].value_counts())

df['Source'].value_counts().plot(
    kind='bar',
    figsize=(8,5)
)

plt.title("Common Information Sources")
plt.xlabel("Sources")
plt.ylabel("Count")
plt.xticks(rotation=45)

plt.show()
```

```
Source
Financial Consultants      16
Newspapers and Magazines   14
Television                  6
Internet                    4
Name: count, dtype: int64
```



Task 8: Investment Duration

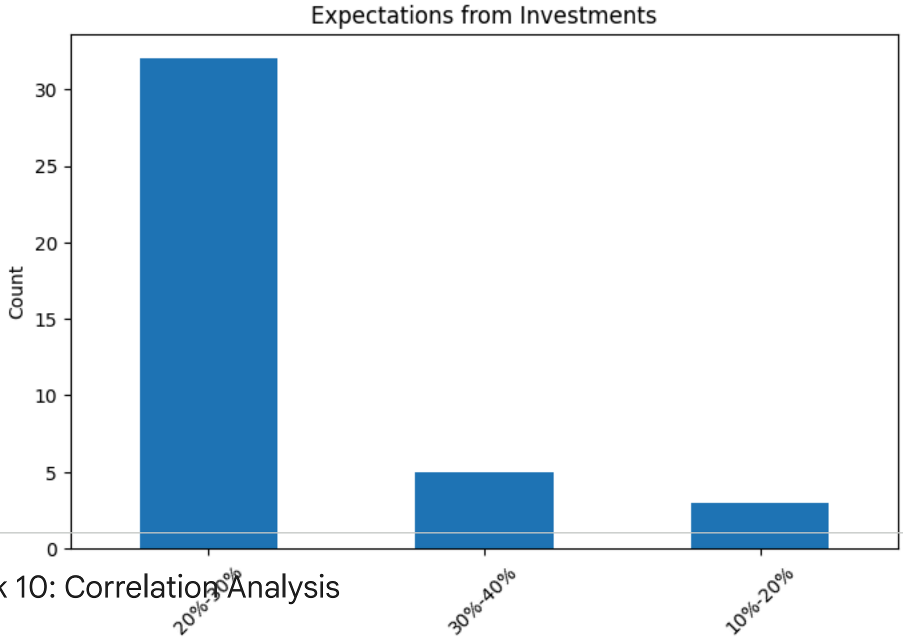
```
#task 8

print(df['Duration'].unique())

mapping = {
    'Less than 1 year': 0.5,
    '1-3 years': 2,
    '3-5 years': 4,
    'More than 5 years': 6
}

df['Duration_Num'] = df['Duration'].map(mapping)

print("Average Investment Duration:")
print(df['Duration_Num'].mean())
```

```
['1-3 years' 'More than 5 years' '3-5 years' 'Less than 1 year']
Average Investment Duration:
2.975
```

Task 9: Expectations from Investments

```
#task 9

print(df['Expect'].value_counts())

df['Expect'].value_counts().plot(
    kind='bar',
    figsize=(8,5)
)

plt.title("Expectations from Investments")
plt.xlabel("Expectations")
plt.ylabel("Count")
plt.xticks(rotation=45)

plt.show()
```

```
Expect
20%-30%    32
30%-40%     5
10%-20%     3
Name: count, dtype: int64
```



Task 10: Correlation Analysis

```
#task 10

corr_columns = [
    'age',
    'Mutual_Funds',
    'Equity_Market',
    'Government_Bonds',
    'Fixed_Deposits',
    'Gold'
]

corr = df[corr_columns].corr()

print(corr)
```

```
                       age  Mutual_Funds  Equity_Market  Government_Bonds  \
age               1.000000     -0.123914       0.246840         -0.093632
Mutual_Funds     -0.123914      1.000000       0.332043         -0.114198
Equity_Market     0.246840      0.332043       1.000000         -0.237420
Government_Bonds -0.093632     -0.114198      -0.237420          1.000000
Fixed_Deposits   -0.033685     -0.031604      -0.238705         -0.531359
Gold             -0.057952     -0.401830      -0.050027         -0.300607

                  Fixed_Deposits      Gold
age                    -0.033685 -0.057952
Mutual_Funds           -0.031604 -0.401830
Equity_Market          -0.238705 -0.050027
Government_Bonds       -0.531359 -0.300607
Fixed_Deposits          1.000000 -0.092730
Gold                   -0.092730  1.000000
```

```
plt.figure(figsize=(8,6))

sns.heatmap(
    corr,
    annot=True,
    cmap='coolwarm'
)

plt.title("Correlation Analysis")

plt.show()


plt.figure(figsize=(7,5))

plt.scatter(
    df['age'],
    df['Mutual_Funds']
)

plt.xlabel("Age")
```

```
plt.ylabel("Mutual Funds")
plt.title("Age vs Mutual Funds")

plt.show()
```